

ICS 4111: Embedded Systems & IoT

Practical Exercise 3: Embedded System Design Lab Report



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Experiment 1: Resistor Value Determination & Multimeter Measurement

1. Nominal Resistance (Colour Code)

Resistor Colour Bands: [Blue, Grey, Brown, Gold]

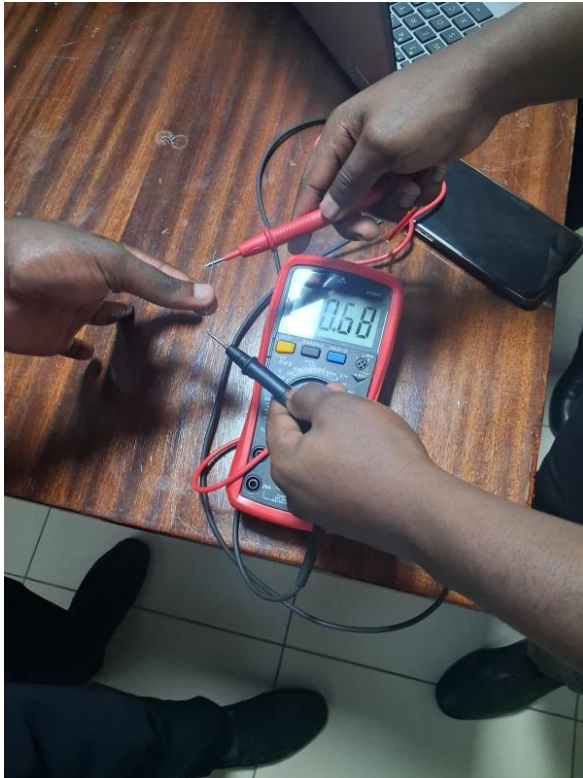
Decoded Resistance Value: 680Ω



2. Measured Resistance (Multimeter)

Multimeter Dial Setting: Resistance mode: it was set to the Kilohms Mode

Measured Value: $0.68 = 680\Omega$



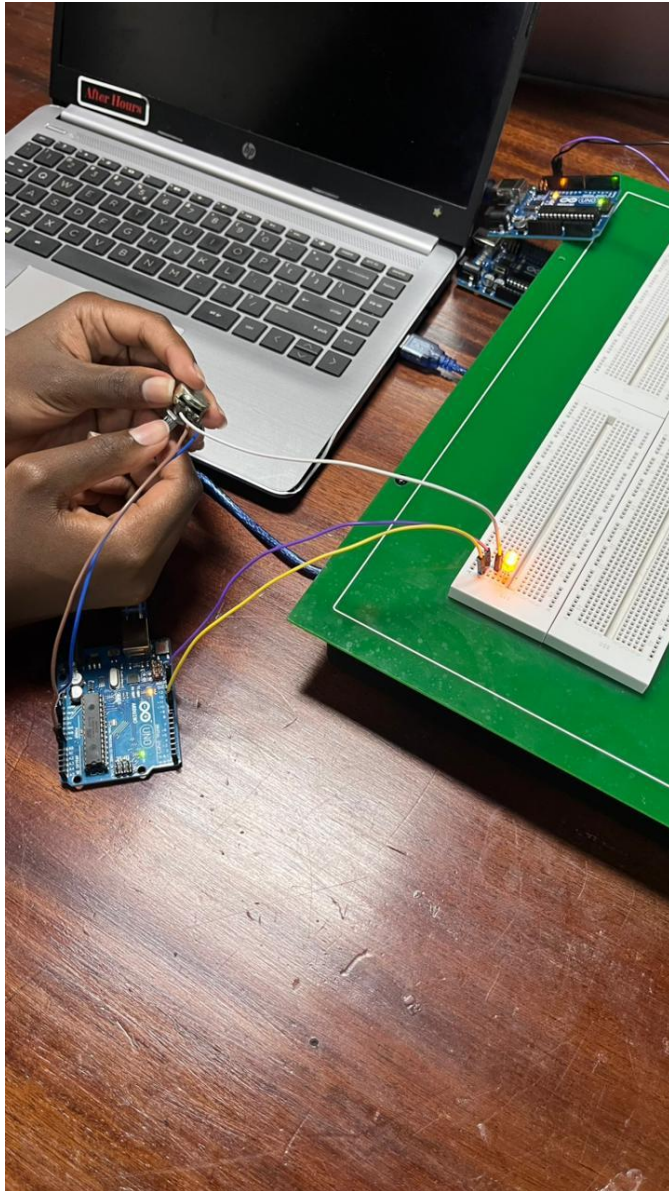
Experiment 1 Reflection: Where does the difference in the resistance values arise from?

The small difference between the resistor's colour code value could come down to standard manufacturing tolerance. Resistors aren't made to be absolutely perfect; the gold band indicates a 5% acceptable margin of error from the factory. In our case, the reading is spot on, showing the resistor is working exactly as it should.

Experiment 2: LED Blinking & PWM Intensity Control

1 & 2. Physical Implementation & Code Deployment

We successfully wired the microcontroller, the 680 Ω resistor, and the LED on the breadboard. The standard built-in Blink sketch was uploaded via the Arduino IDE, which successfully toggled the LED on and off at 1-second intervals.



3. Voltage Measurement

Using the multimeter set to the DC Voltage V = range, we measured the voltage across the LED circuit while the pin was in the HIGH state.

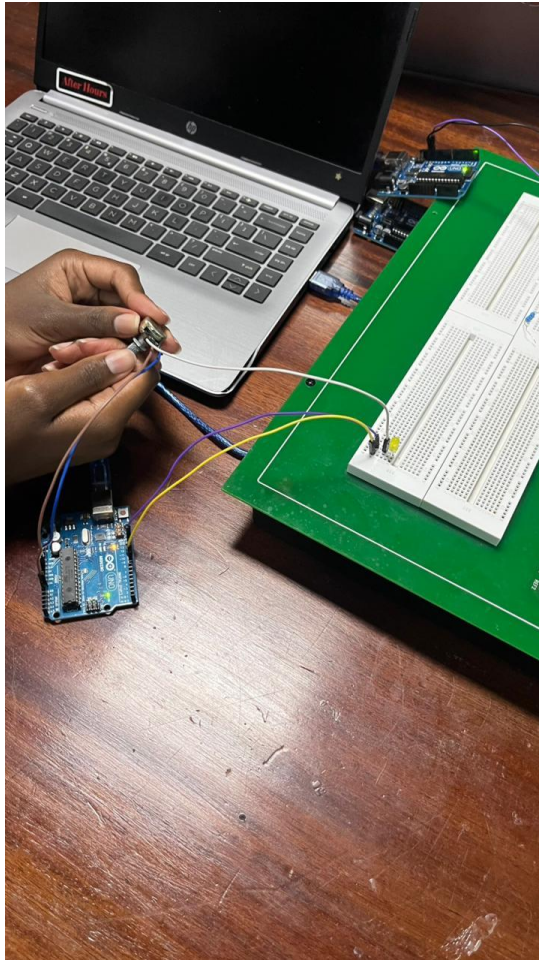


4. PWM Dimming with Potentiometer

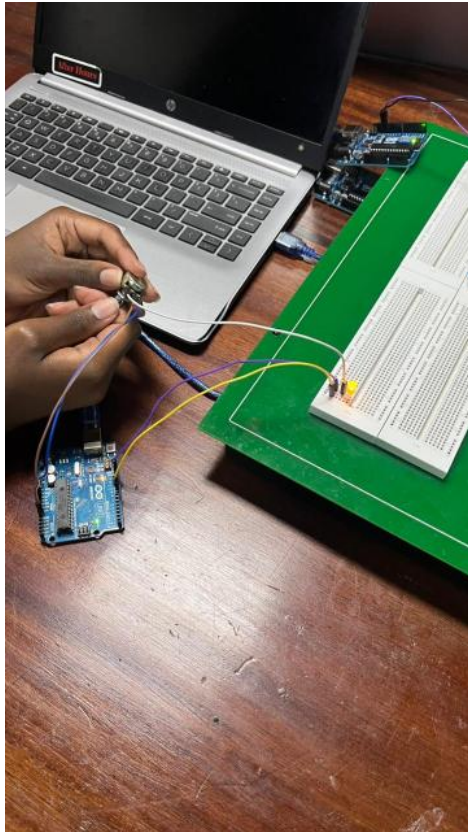
We reconfigured the circuit by introducing a potentiometer to control the LED's brightness. The potentiometer's middle pin was connected to an analog input pin, and its value was mapped to a PWM-enabled output pin to adjust the duty cycle.

We captured the circuit at three distinct brightness levels:

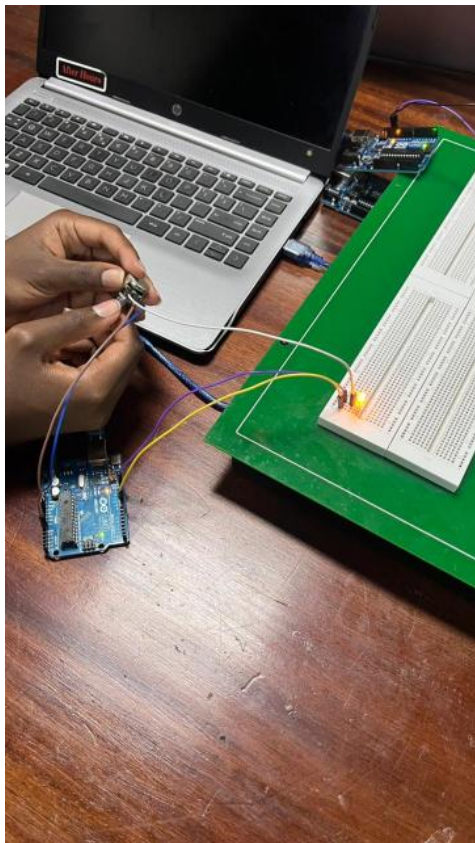
- **25% Brightness:** The LED is dimly lit. The microcontroller sends rapid voltage pulses with a short "on" time (25% duty cycle)



- **50% Brightness:** The LED is at medium intensity. The pulses have equal "on" and "off" time (50% duty cycle).



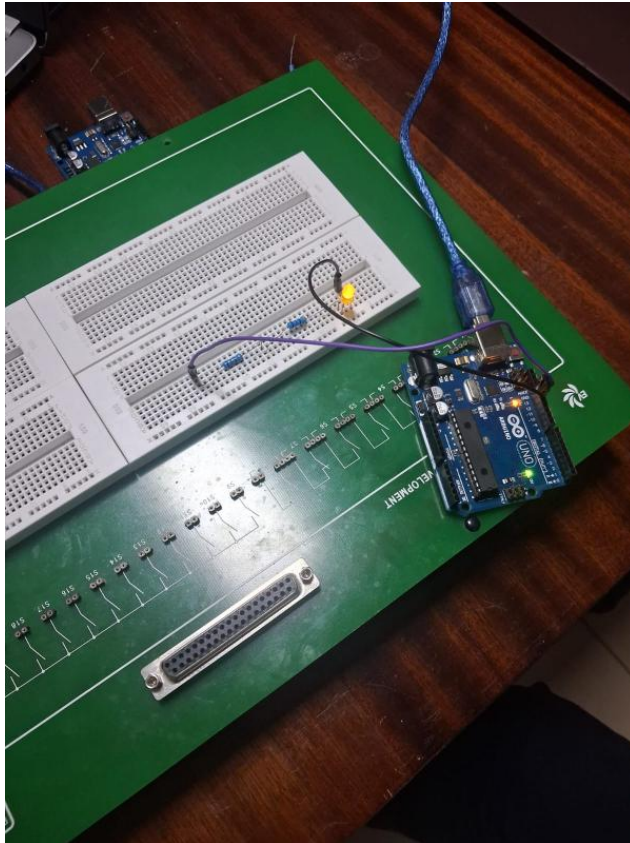
- **100% Brightness:** The LED is at maximum intensity. The microcontroller provides a constant, steady voltage



Experiment 3: Resistors in Series

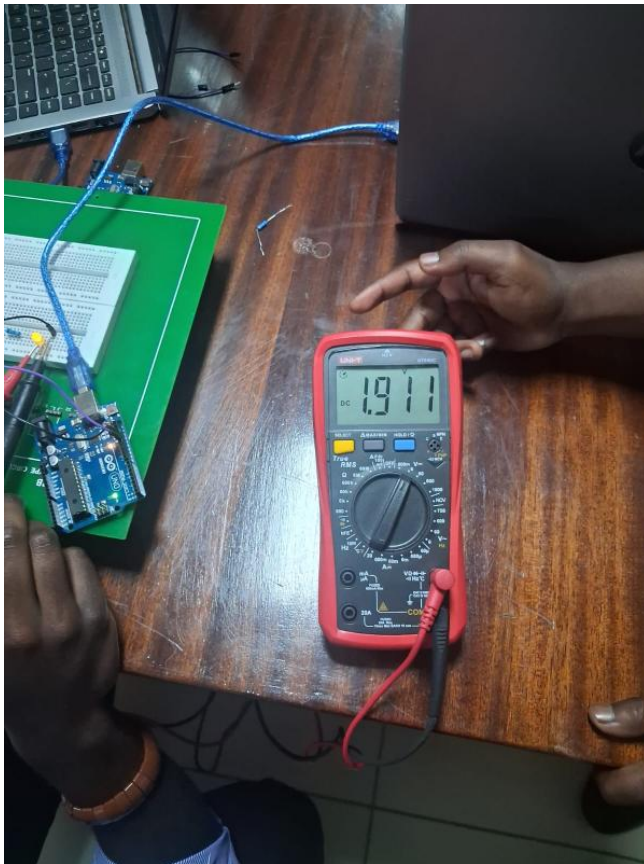
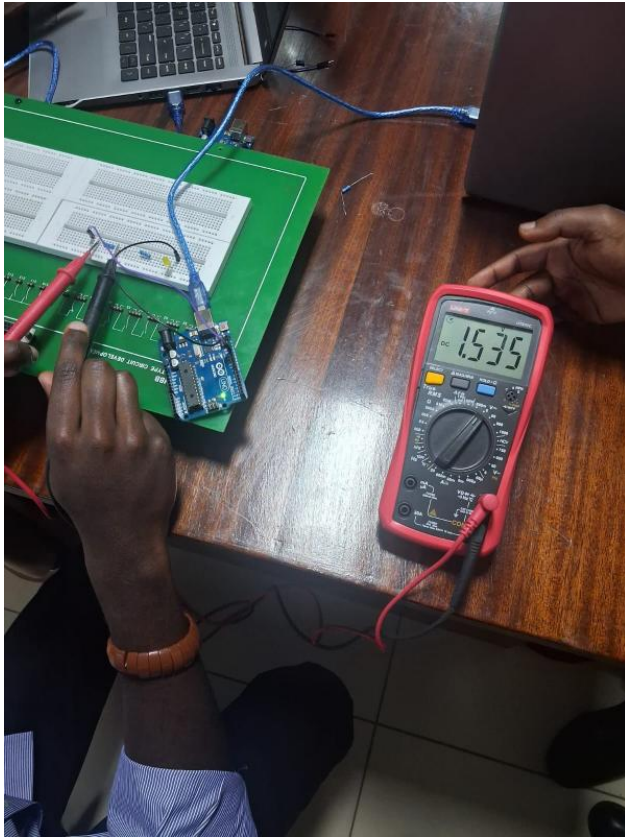
1. Physical Implementation & Code Deployment

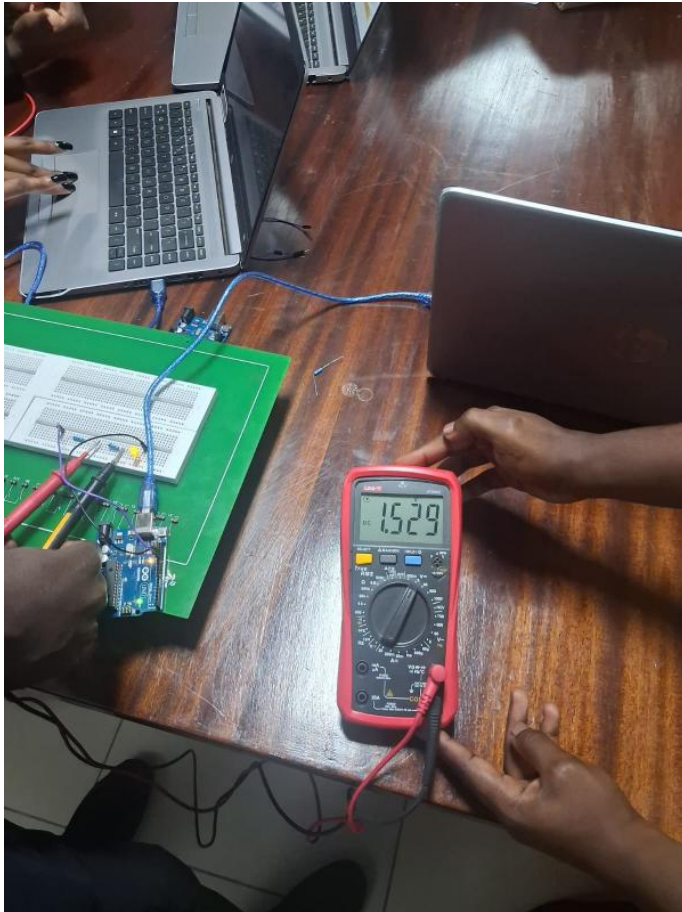
Two $680\ \Omega$ resistors were connected in a series configuration together with the LED and the microcontroller's digital output pin. The standard Blink program was uploaded and executed to periodically power the circuit.



2. Voltage Measurements

Using a digital multimeter, the voltage drop across each component was measured while the circuit was operating.





Component	Measured Voltage
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Resistor 1 (V_{R1})	1.535 V
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Resistor 2 (V_{R2})	1.529 V
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LED (V_{LED})	1.911 V
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Verification of Kirchhoff's Voltage :

$$\begin{aligned}V_{Total} &= V_{R1} + V_{R2} + V_{LED} \\V_{Total} &= 1.535 + 1.529 + 1.911 \\V_{Total} &= 4.975 \text{ V}\end{aligned}$$

The total measured voltage drop of 4.975 V closely matches the microcontroller's 5 V supply voltage, confirming that the measurements are consistent with circuit theory.

3. Current Calculations Using Ohm's Law

In a series circuit, the current is the same through all components:

$$I_{Total} = I_{R1} = I_{R2} = I_{LED}$$

Using Ohm's Law:

$$I = \frac{V}{R}$$

where:

- V = measured voltage across the resistor
- $R = 680 \, \Omega$

Current Through Resistor 1

$$\begin{aligned} I_{R1} &= \frac{V_{R1}}{R} \\ I_{R1} &= \frac{1.535}{680} \\ I_{R1} &= 0.002257 \text{ A} \\ I_{R1} &= 2.26 \text{ mA} \end{aligned}$$

Current Through Resistor 2

$$\begin{aligned} I_{R2} &= \frac{V_{R2}}{R} \\ I_{R2} &= \frac{1.529}{680} \\ I_{R2} &= 0.002249 \text{ A} \\ I_{R2} &= 2.25 \text{ mA} \end{aligned}$$

Average Circuit Current

$$\begin{aligned} I_{Average} &= \frac{I_{R1} + I_{R2}}{2} \\ I_{Average} &= \frac{2.26 + 2.25}{2} \\ I_{Average} &= 2.255 \text{ mA} \\ I_{Average} &= 2.26 \text{ mA} \end{aligned}$$

4. Reflection and Analysis

The calculated currents through both resistors are nearly identical:

$$\begin{aligned} I_{R1} &= 2.26 \text{ mA} \\ I_{R2} &= 2.25 \text{ mA} \end{aligned}$$

The small difference can be attributed to normal measurement tolerances. These results confirm that the same current flows through every component in the circuit.

The equivalent resistance of the two series resistors is:

$$R_{Total} = R_1 + R_2$$

$$R_{Total} = 680\ \Omega + 680\ \Omega$$

$$R_{Total} = 1360\ \Omega$$

Doubling the resistance reduced the current compared to the single-resistor in experiment 2. As a result, less current flowed through the LED, causing it to emit less light and appear noticeably dimmer. This observation is consistent with Ohm's Law, which states that current decreases as resistance increases when the supply voltage remains constant.

Experiment 4: Resistors in Parallel

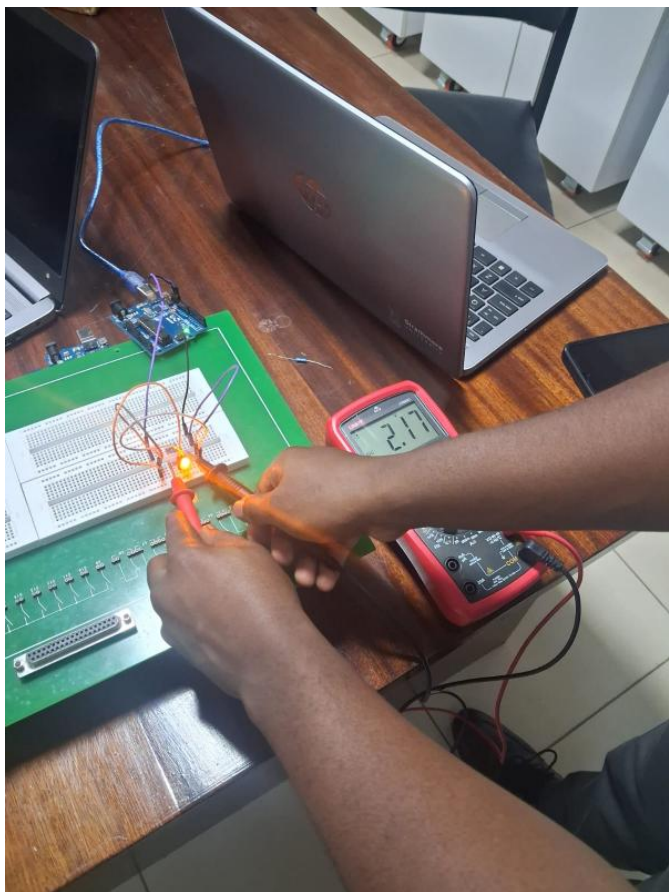
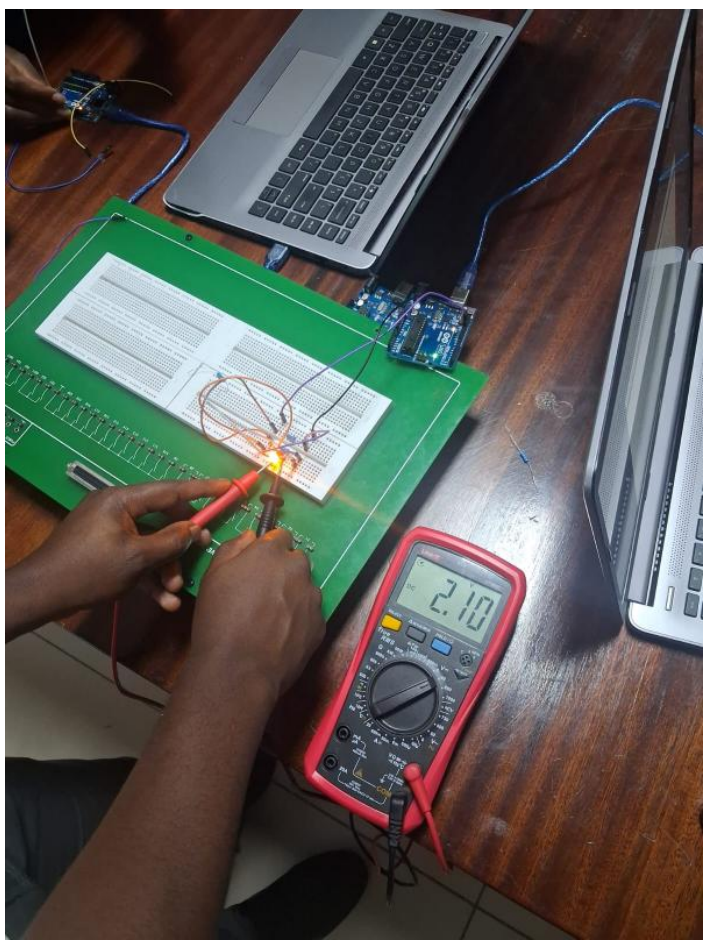
1. Physical Implementation & Code Deployment

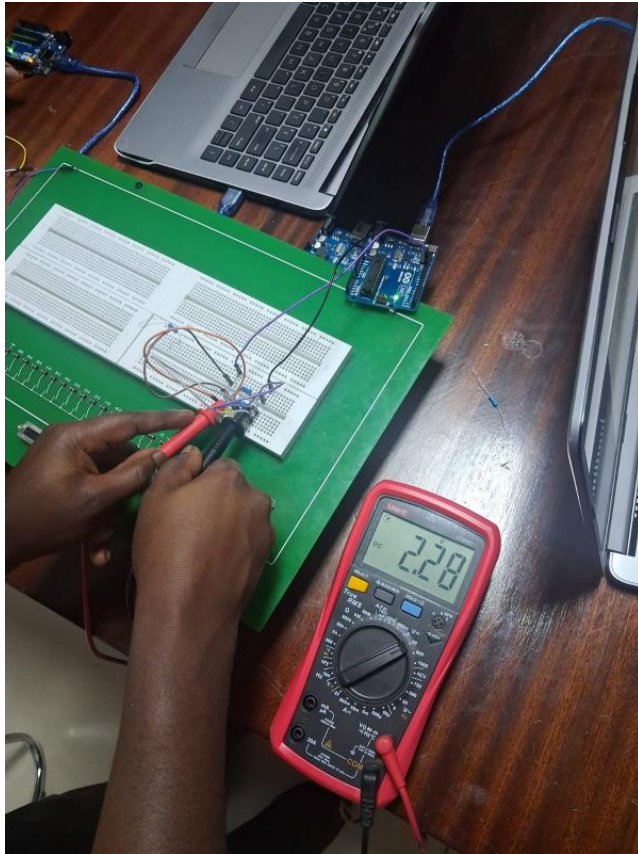
Two $680\ \Omega$ resistors were connected in a parallel configuration, meaning both resistors shared the same input and output nodes. The parallel resistor network was then connected in series with the LED and the microcontroller output pin. The standard Blink program was uploaded and executed to periodically power the circuit.



2. Voltage Measurements

Using a digital multimeter, the voltage drop across each component was measured while the circuit was operating.





Component	Measured Voltage
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Resistor 1 (V_{R1})	2.10 V
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Resistor 2 (V_{R2})	2.17 V
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LED (V_{LED})	2.28 V
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Since the resistors are connected in parallel, they should ideally have the same voltage across them. The small difference between the measured values is due to normal measurement uncertainty.

Average Voltage Across the Parallel Network

$$V_{Parallel} = \frac{V_{R1} + V_{R2}}{2}$$
$$V_{Parallel} = \frac{2.10 + 2.17}{2}$$
$$V_{Parallel} = 2.135 \text{ V}$$
$$V_{Parallel} = 2.14 \text{ V}$$

Total Measured Voltage Drop

$$V_{Total} = V_{Parallel} + V_{LED}$$

$$V_{Total} = 2.14 + 2.28$$

$$V_{Total} = 4.42 \text{ V}$$

The measured voltage is slightly lower than the nominal 5 V supply voltage. The remaining voltage is attributed to internal voltage drops within the microcontroller.

3. Current Calculations Using Ohm's Law

Ohm's Law states:

$$I = \frac{V}{R}$$

where:

- V = measured voltage across the resistor
- $R = 680 \, \Omega$

In a parallel circuit, the total current is the sum of the currents flowing through each branch:

$$I_{Total} = I_{R1} + I_{R2}$$

Current Through Resistor 1

$$I_{R1} = \frac{V_{R1}}{R}$$

$$I_{R1} = \frac{2.10}{680}$$

$$I_{R1} = 0.003088 \text{ A}$$

$$I_{R1} = 3.09 \text{ mA}$$

Current Through Resistor 2

$$I_{R2} = \frac{V_{R2}}{R}$$

$$I_{R2} = \frac{2.17}{680}$$

$$I_{R2} = 0.003191 \text{ A}$$

$$I_{R2} = 3.19 \text{ mA}$$

Total Current Through the LED

$$I_{Total} = I_{R1} + I_{R2}$$

$$I_{Total} = 3.09 \text{ mA} + 3.19 \text{ mA}$$

$$I_{Total} = 6.28 \text{ mA}$$

$$I_{Total} = 0.00628 \text{ A}$$

4. Equivalent Resistance of the Parallel Network

For two resistors connected in parallel:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

$$R_{eq} = \frac{680 \times 680}{680 + 680}$$

$$R_{eq} = \frac{462400}{1360}$$

$$R_{eq} = 340 \Omega$$

5. Reflection and Analysis

The calculated branch currents were:

$$I_{R1} = 3.09 \text{ mA}$$

$$I_{R2} = 3.19 \text{ mA}$$

Combining these currents gives a total circuit current of:

$$I_{Total} = 6.28 \text{ mA}$$

The equivalent resistance of the parallel resistor network was 340Ω , which is half the resistance of a single 680Ω resistor. Because the total resistance was significantly reduced compared to the series configuration in Experiment 3 (1360Ω), a much larger current was able to flow through the circuit.

Comparing the two experiments:

Configuration Equivalent Resistance Current

Series	1360Ω	2.26 mA
Parallel	340Ω	6.28 mA

The parallel circuit allowed nearly three times more current to flow than the series circuit. Thus, the LED received more current and emitted a much brighter light. This observation agrees with Ohm's Law, which predicts that reducing resistance while maintaining approximately the same supply voltage increases the current flowing through the circuit.